Proof:-

Let $T_1, T_2 \in L(U,V)$

$$(T_1+T_2)(\alpha u_1+\beta u_2) = T_1(\alpha u_1+\beta u_2) + T_2(\alpha u_1+\beta u_2)$$
$$= \alpha T_1(u_1) + \beta T_1(u_2) + \alpha T_2(u_1) + \beta T_2(u_2)$$
$$= \alpha(T_1+T_2)(u_1) + \beta(T_1+T_2)(u_2)$$

$\therefore T_1+T_2 \in L(U,V)$.

Q:- $R(T) = \{(x_1,x_2,x_3) \in V_3 \mid 4x_1-3x_2+x_3=0\}$

Find T.

Q:- $T: \mathbb{R}^2 \to \mathbb{R}^2$

$T(x,y) = (2x, 3y)$

S: $x^2+y^2=1$

$T(S) = ?$

$T^{-1}(S) = ?$

$T(1,0) = (2,0)$

$1+0=1$

$(2,3)$

A:- Let $T(x,y) = (s,t)$

$\Rightarrow (s,t) = (2x,3y)$

$\Rightarrow s = 2x, \quad t = 3y$

$\Rightarrow x = \frac{s}{2}, \quad y = \frac{t}{3}$.

$T(S): \left(\frac{s}{2}\right)^2 + \left(\frac{t}{3}\right)^2 = 1$

$\therefore \frac{s^2}{4} + \frac{t^2}{9} = 1$

$T^{-1}(S): (2x)^2 + (3y)^2 = 1$

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